



北京大學
PEKING UNIVERSITY

Small Molecule on Ice Surface

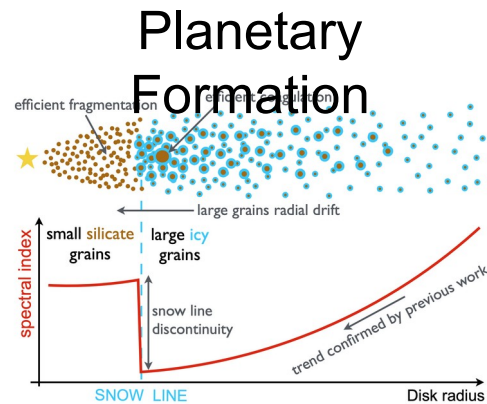
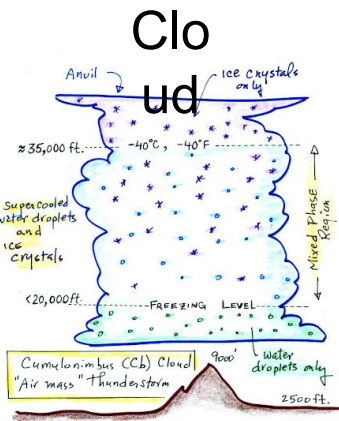
Group Meeting – 2025/03/13

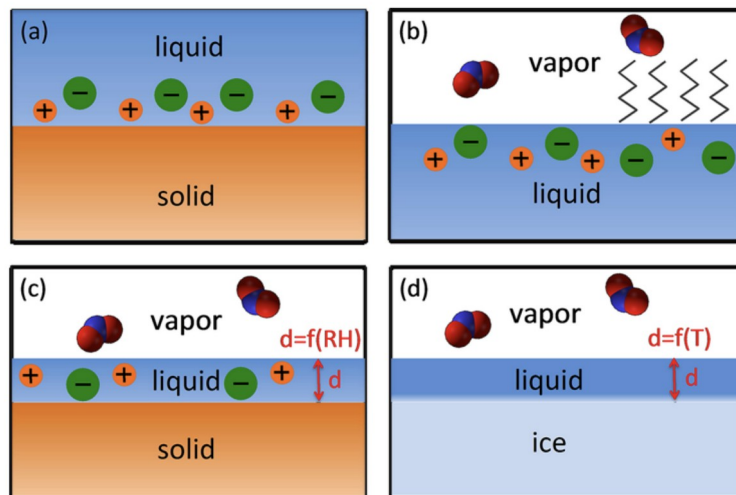
Presenter: Chon-Hei Lo

Ice surface appears in different environments and plays a crucial role in many fields.

Formation of ice surface could affect 相互作用的形式。 It affects how ice crystal grows in cloud, molecule abundance in diffuse cloud, design of anti-freezing battery.

More specifically, friction, adhesion, adsorption, reaction, and release of trace gases...





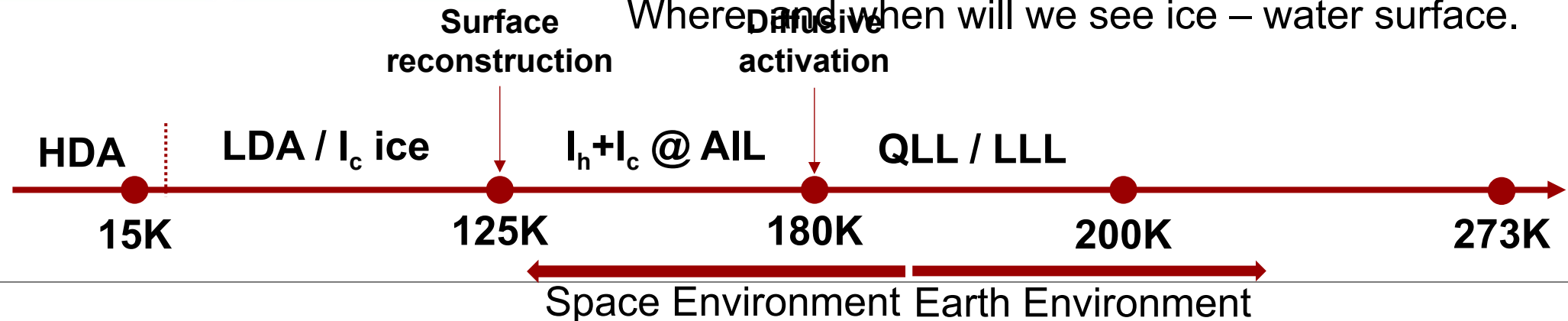
There are many types of ice interface that scientist interest:

1. metal @ liquid water
2. Surface of pure liquid water
3. Thin water layer on metal surface
4. Ice – water surface.

In the talk, we focus on ice – water surface.

The first thing we need to classify:

Where and when will we see ice – water surface.



Behavior on surface including **adsorption**, **desorption**, **diffusion**, (and **chemical reaction**).

1. Adsorption & Desorption are controlled by QM and related to protons.

Many astronomers would approximate the diffusive behavior as $E_{\text{diff}} = f E_{\text{bind}}$, $f \sim 0.3$.

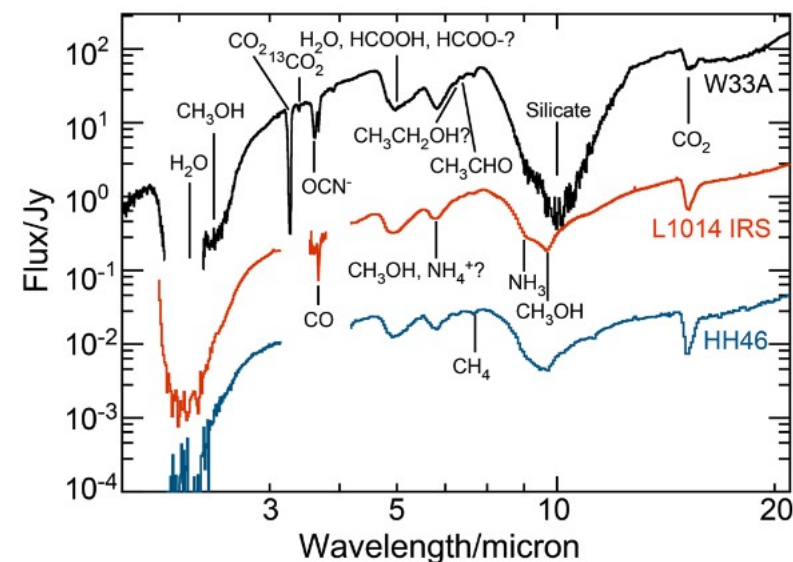
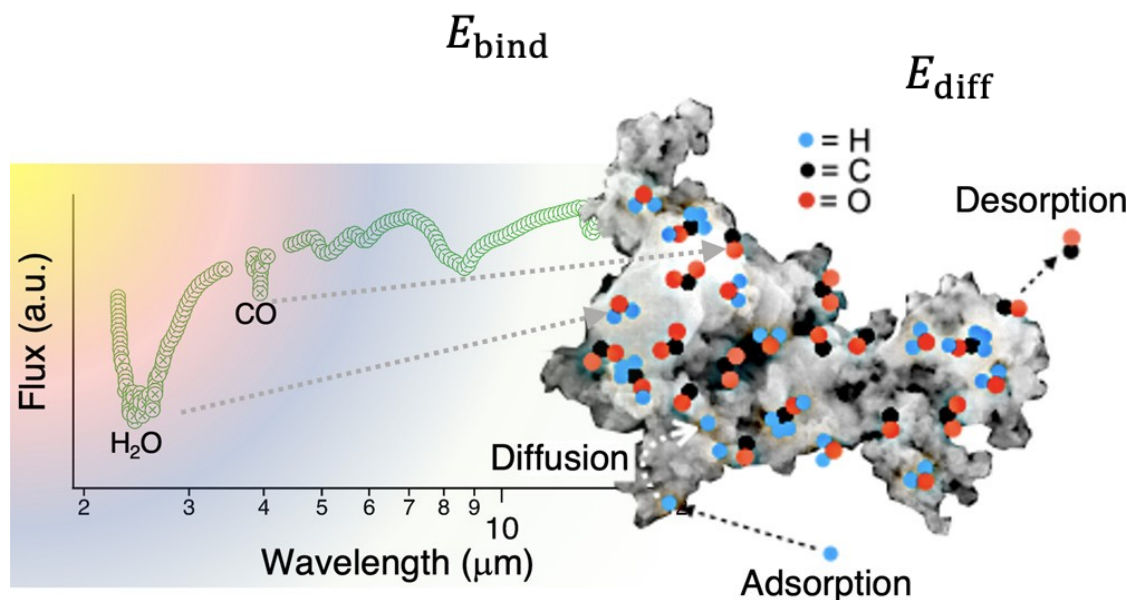
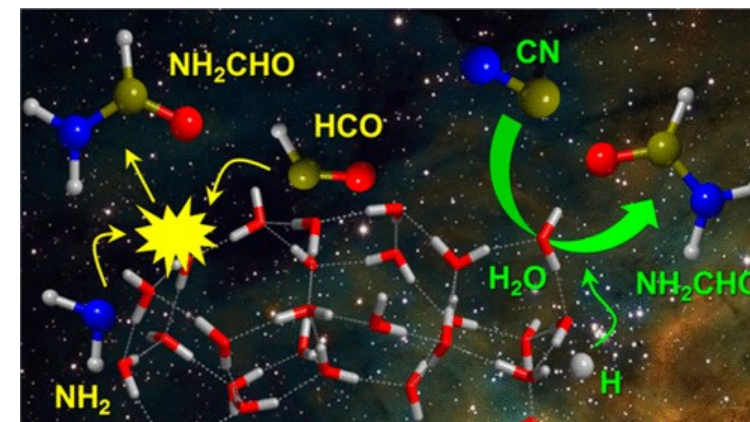
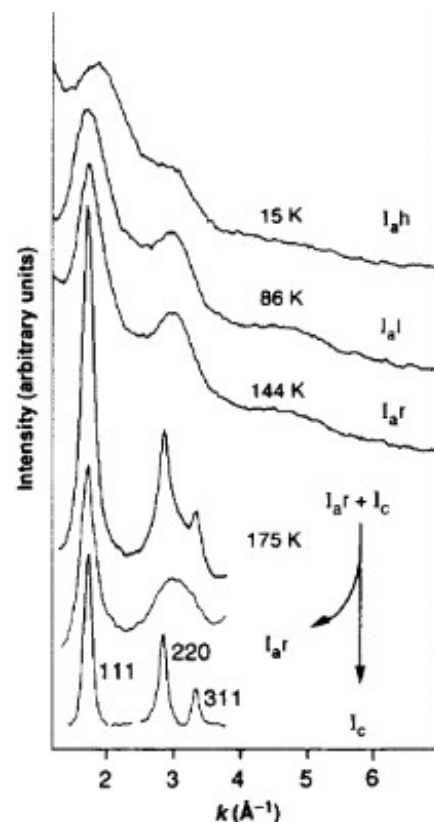
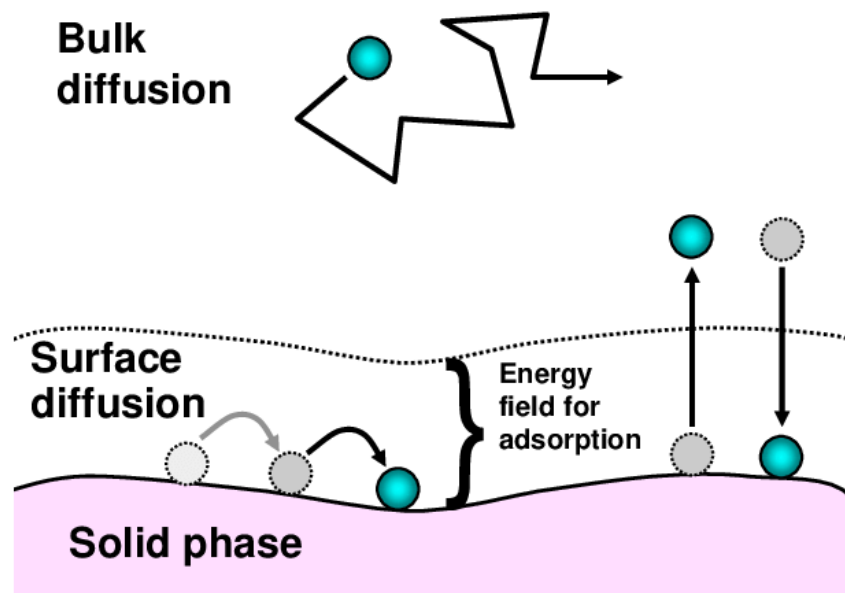


Figure 3. Spectra of interstellar ice absorptions seen toward 3 protostellar objects. Spectrum taken from Öberg et al.¹⁶ Copyright 2011 IOP Publishing.

Interstellar Ice

Ice was found in the molecular clouds and comets last century.

They exist in different forms and E_{bind} varies.

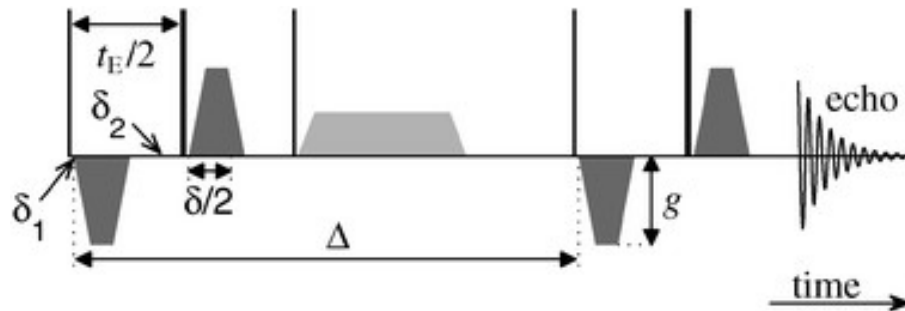


0–80 K	80–120 K	120–160 K	>160 K
porous amorphous	compact amorphous	crystalline	gas
HDA	LDA	I_c (cubic)	
p-ASW	np-ASW	I_h (hexagonal)	
	c-ASW	CSW	

Measurement of Diffusibility on Surface

Indirect Method – PFG NMR

1. Apply magnetic gradient pulse to label the atoms, yielding an NMR signal S_i .
2. Relax a long period Δ .
3. Repeat (1), yielding an echo signal S_f .
4. The diffusion coefficient is $D = \frac{1}{(\gamma g \delta)^2} \frac{1}{\Delta} \ln \frac{S_i}{S_f}$



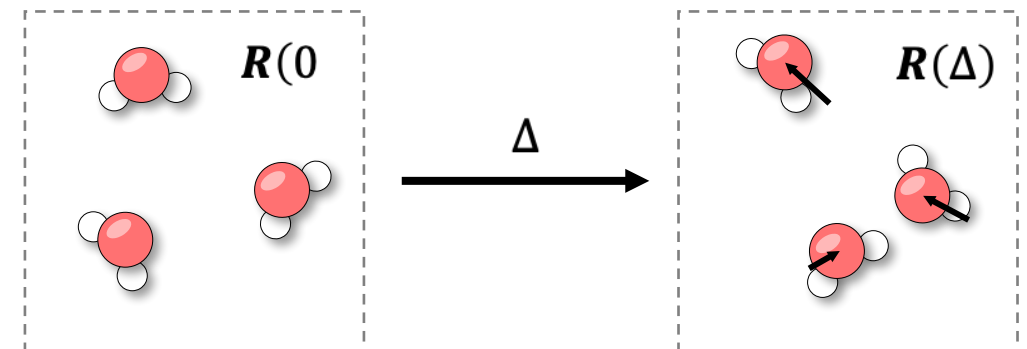
*Pulsed Field Gradient Nuclear Magnetic Resonance

Direct Method – Microscopy

e.g. STM, AFM, FIM, ...

Estimate from the definition:

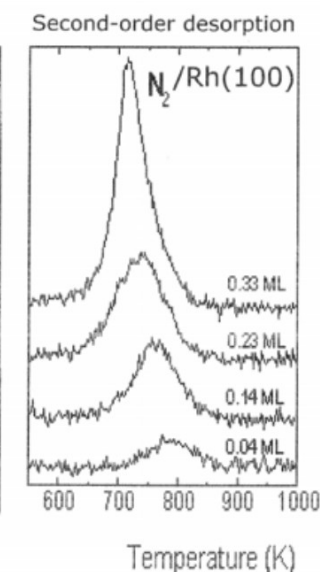
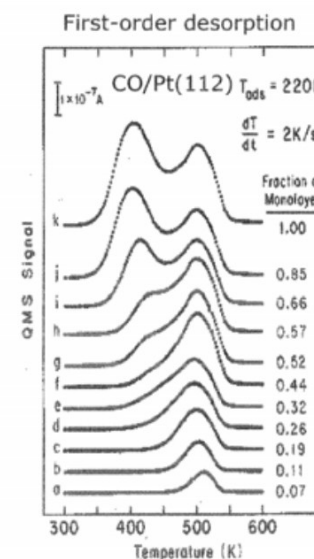
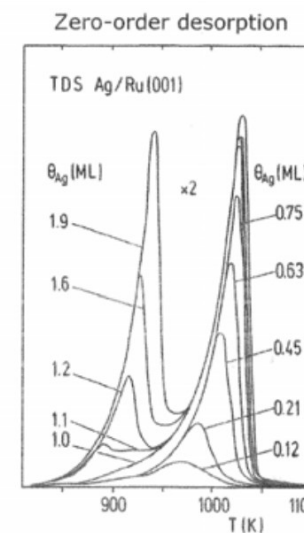
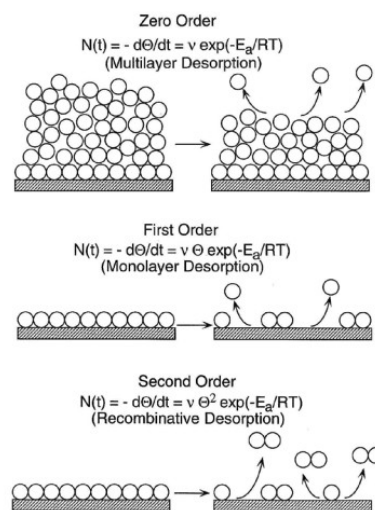
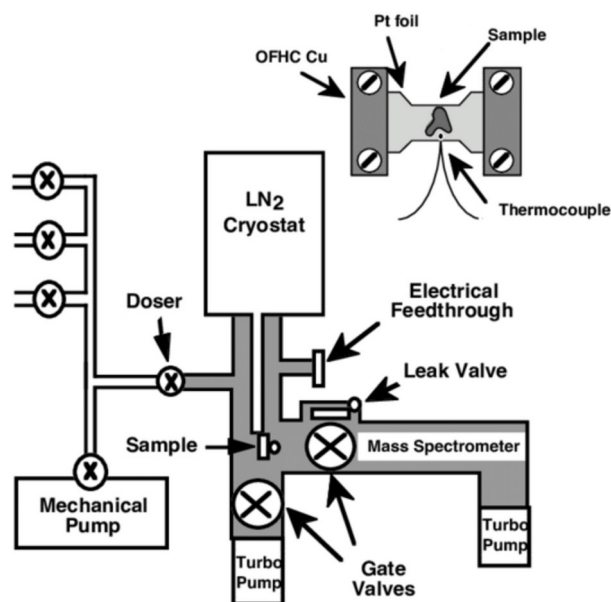
$$D \approx \frac{1}{2d\Delta} \langle |\mathbf{R}(\Delta) - \mathbf{R}(0)|^2 \rangle.$$



Measurement of Binding Energy

TPD: coverage Θ , measuring $\frac{d\Theta}{dt} \propto e^{-\frac{E_{\text{bind}}}{k_b T}}$ is equivalent to measure binding energy E_{bind} .

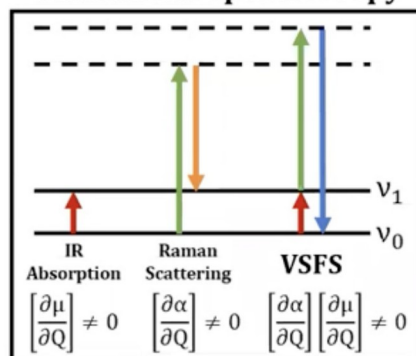
Step: heating up the sample “very slowly” (~hrs), using mass spectrometer to identify the type of molecules.



*Temperature-Programmed Desorption (TPD)

Vibrational Sum Frequency Spectroscopy (VSFS)

Vibrational Spectroscopy



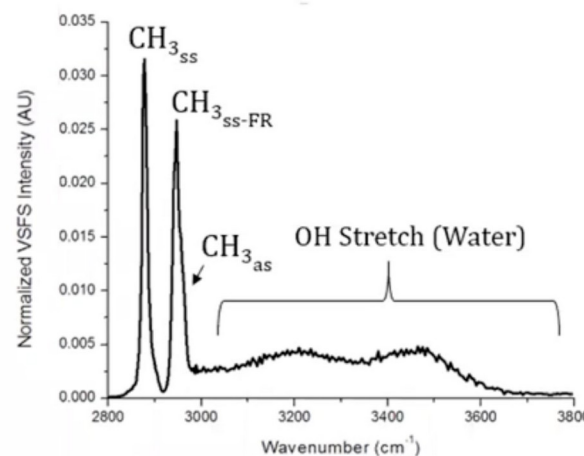
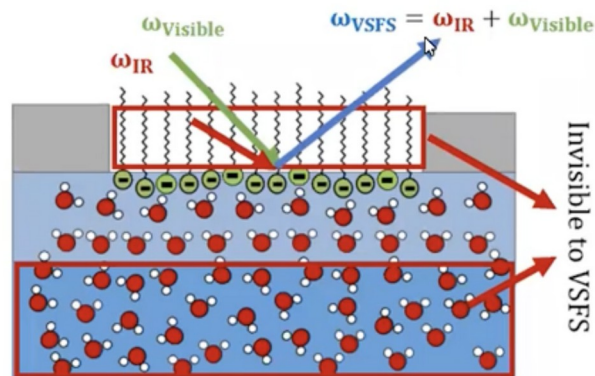
$\left[\frac{\partial \mu}{\partial Q}\right]$ Change in dipole

$\left[\frac{\partial \alpha}{\partial Q}\right]$ Change in polarizability

*Selection rules result in no signal for molecular vibrations in a centrosymmetric environment



K Judd



Others: VSFS, IR, QCM, XPS, ...

How to determine the binding energy?

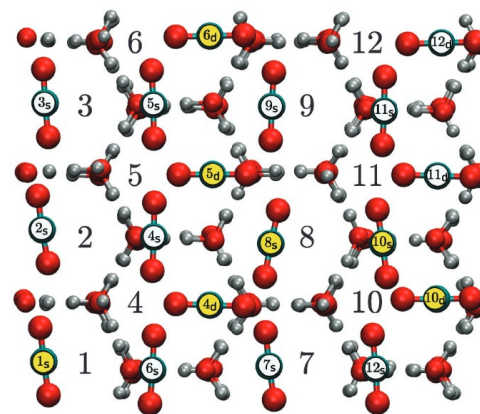
E_{bind} of one **molecule** depends on **structure**, **dangling OH-bonds**, and **coverage**.

E_{bind} is also **multivalued** when changing the sites.

Determine structure

Find binding sites

symmetry

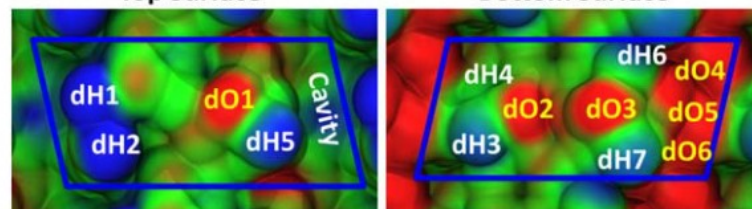


HF
DFT
KMC
GCMC

electric potential map

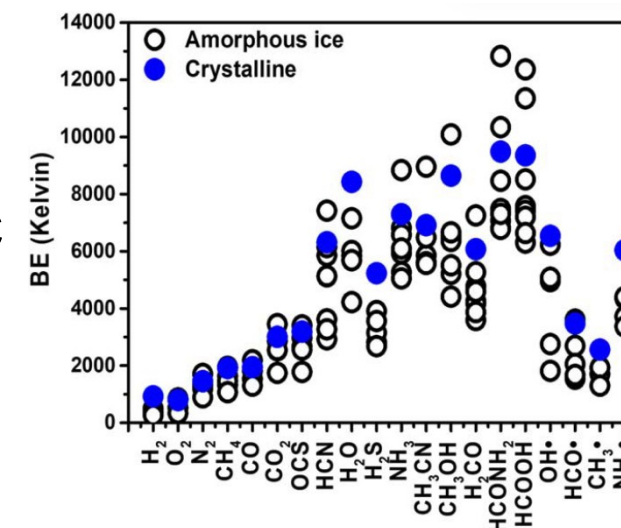
Top Surface

Bottom Surface



HF-3c

Analysis BE!



*binding energy
*1000K ~ 0.09eV

BE of Water Molecule on Ice I_h

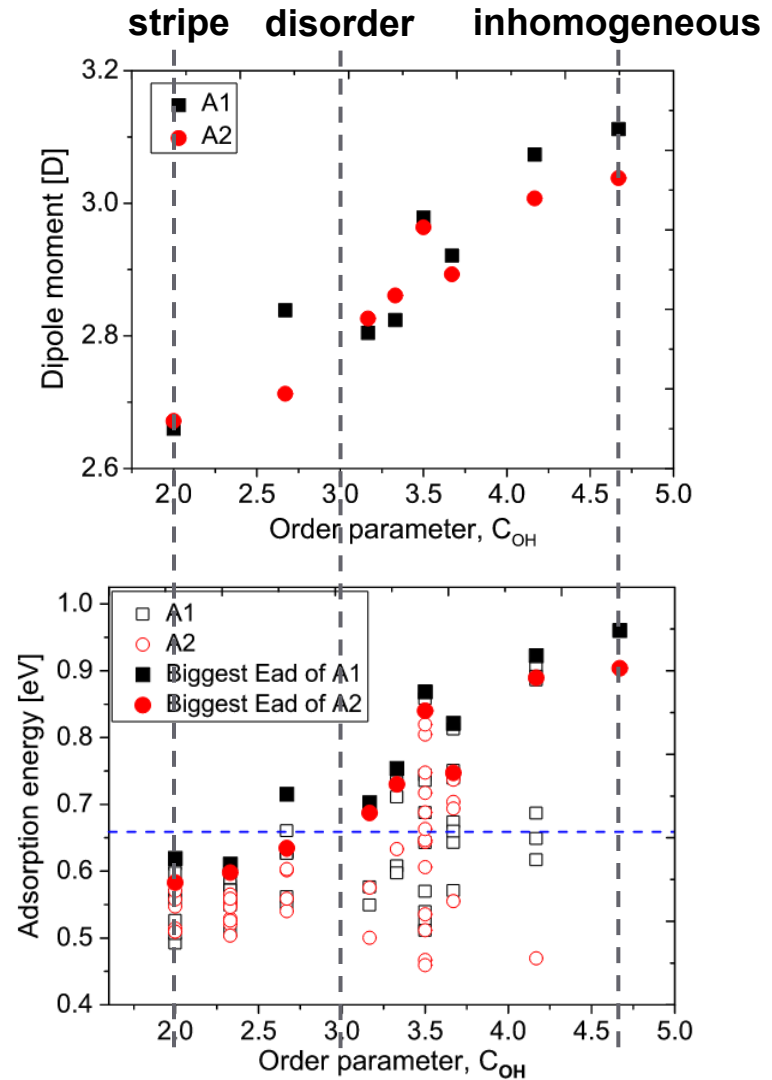
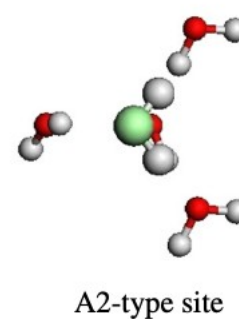
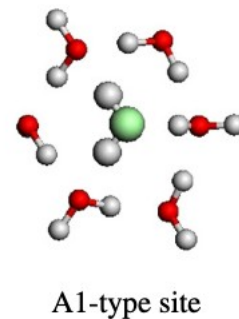
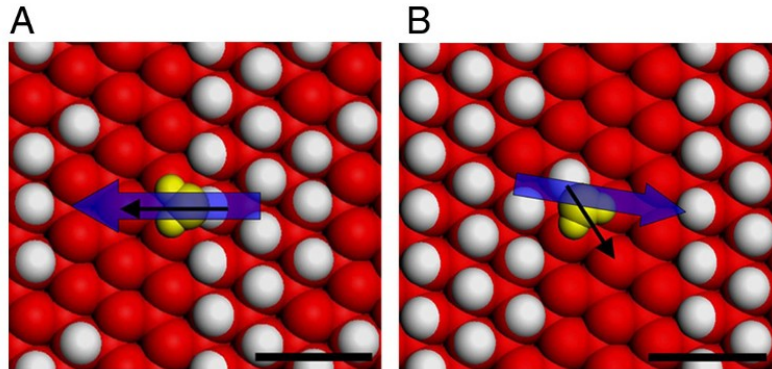
Proton order affects BE - Define proton order

$$C_{OH} = \frac{1}{N_{OH}} \sum_{i < j} \delta_{ij},$$

where δ_{ij} means neighbor dangling bonds pair.

Generate structure by C_{OH}

Find binding sites



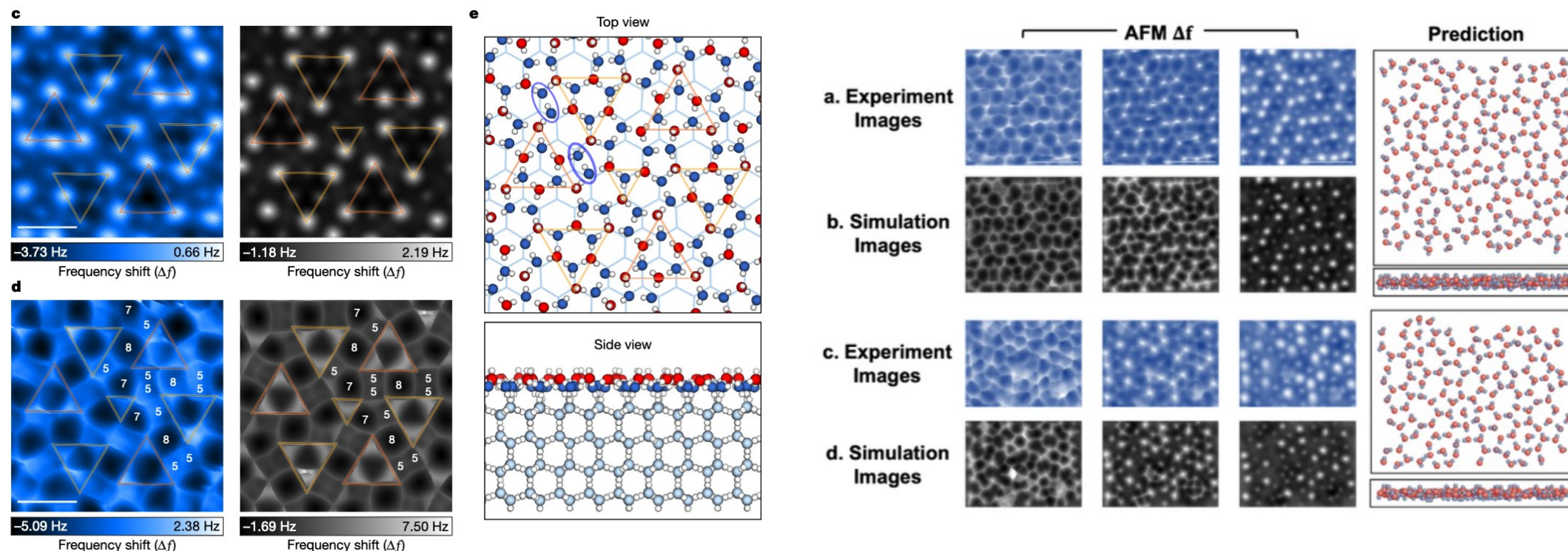
Vacancy Formation Energy of Water Molecule on Ice I_h

VFE is another form of BE when coverage Θ is big. In ice, it also varies with the local electrostatic field, and the polarization of leaving H_2O molecule.

VFE varies when $\Theta \rightarrow 1^-, 2^-, 3^-, \dots$

What if Ice Surface Encounters Premelting?

Our study indicates at 120K, ice surface will **reconstruct**! The AIL exists <180K also changes the game rule – all calculation at >120K does not reflect the real situation.



**Amorphous ice layer*

What if Ice Surface Encounters Premelting?

1. The **dipole moment distribution** will be denser?
2. The **vacancy formation energy** is higher? → the reactivity may change?
3. How different molecules **bind** and **diffuse** on this surface? And correlated with proton order?
4. Could we estimate the abundance of organic molecule more accurately – Of course, we can try to resolve some **unidentified infrared emission bands**.